

WEEK 1: WAZUH SIEM LAB DEPLOYMENT REPORT

Executive Summary

The objective of this lab was to gain practical experience with Security Information and Event Management (SIEM) technology by deploying a Wazuh monitoring environment. The deployment consisted of a Wazuh Manager running as a virtual appliance on Oracle VirtualBox and a Wazuh Agent installed on a Kali Linux virtual machine. The main goal was to establish successful communication between the manager and the agent and verify that security events were being collected and processed correctly.

This setup represents a simplified version of a real-world security monitoring environment where endpoint agents continuously send system and security data to a centralized platform for analysis. Successful deployment was confirmed when the Wazuh Dashboard displayed "1 Agent Active," indicating that the agent had been successfully enrolled and was communicating with the manager. The completed environment provides a foundation for future activities such as log monitoring, file integrity monitoring, vulnerability detection, and security analysis.

Methodology

The deployment process began with the installation of Oracle VirtualBox and the creation of the virtual machines required for the lab environment. The Wazuh Manager virtual machine was allocated approximately four CPU cores, eight gigabytes of RAM, and fifty gigabytes of storage to ensure adequate performance for the manager, indexer, and dashboard components. The network adapter was configured in Bridged Mode to allow direct communication between the virtual machines.

The official Wazuh OVA appliance was downloaded and imported into VirtualBox. After the initial boot process, a static IP address was configured to maintain consistent network connectivity. The default administrative credentials were changed, and the Wazuh Dashboard was accessed through a web browser to verify that the manager, indexer, and dashboard services were operating correctly.

The Kali Linux virtual machine was then prepared for agent installation. The Wazuh repository and GPG key were added to the system, and the Wazuh Agent package was installed using the APT package manager. During installation, the IP address of the Wazuh Manager was specified so that the agent could communicate with the manager.

once the service was started. After installation, the agent service was enabled and verified to ensure it was running successfully.

Technical Findings

The deployment was completed successfully, and both virtual machines operated without significant issues. The Wazuh OVA appliance booted correctly, and all core services were available after startup. Network communication between the Wazuh Manager and the Kali Linux agent functioned as expected, allowing the enrollment process to complete successfully.

After enrollment, the agent appeared in the dashboard with its assigned agent ID, hostname, operating system information, and IP address. The dashboard reported one active agent and no disconnected or pending agents. The enrolled system was identified as agent11 running Kali GNU/Linux 2026.2 with Wazuh version 4.14.5.

Security events and inventory information began appearing in the dashboard shortly after enrollment. The File Integrity Monitoring, Log Analysis, and Vulnerability Detection modules automatically initiated baseline monitoring activities on the Kali Linux system. The dashboard also showed active alert generation, including low and medium severity alerts collected during the monitoring period. This confirmed that the complete data flow from the agent to the manager, indexer, and dashboard was functioning correctly. The manager console confirms that the Wazuh services are operational, while the dashboard displays one active endpoint and ongoing event processing. These results demonstrate that the SIEM environment was configured successfully and is capable of monitoring endpoint activity.

Recommendations

Several improvements can be made to further strengthen the environment. Default credentials should be replaced with strong passwords and managed according to a defined password policy. Communication between agents and the manager should be secured using properly configured TLS certificates to improve authentication and data protection. The default Wazuh rules should be reviewed and tuned to reduce unnecessary alerts and improve the accuracy of detections. Additional agents can also be deployed to simulate a larger enterprise environment and evaluate the scalability of the platform. Future enhancements may include integrating Wazuh with ticketing systems, notification platforms, or Security Orchestration, Automation, and Response solutions to support incident response activities. Regular backups or snapshots of the

WEEK 2: LOG MONITORING AND THREAT HUNTING USING WAZUH

Executive Summary

The objective of Week 2 was to move beyond basic agent connectivity and begin analyzing the security data generated by the Kali Linux endpoint. After successfully deploying the Wazuh Manager and Agent, the focus shifted to monitoring system logs, investigating authentication-related events, and understanding how Wazuh organizes security information.

Since the monitored endpoint was running Kali Linux, authentication monitoring relied on Linux system logs, PAM records, and SSH-related events rather than Windows Event IDs. By analyzing the collected logs, it was possible to identify failed login attempts and other security-related activities. This phase provided practical experience with log analysis and threat hunting while demonstrating how Wazuh can centralize and monitor endpoint activity.

Methodology

The monitoring process began by accessing the Threat Hunting module within the Wazuh Dashboard and filtering events generated by the Kali Linux agent. Searches were performed using the agent name to ensure that only data from the monitored endpoint was displayed.

Several queries were used to investigate authentication activity. Authentication failure searches were performed to identify unsuccessful login attempts, while broader authentication-related searches were used to capture additional login events. SSH-related filters were also applied because SSH is one of the primary access methods on Linux systems.

To improve organization, a dedicated group was created for the Kali Linux agent instead of relying solely on the default group. This allows easier management and monitoring as additional agents are added in the future.

The Events section was also reviewed to examine the raw logs collected by Wazuh. This provided visibility into authentication events, rootcheck findings, security configuration assessments, and other endpoint activity.

Technical Findings

The analysis identified ten authentication-related failure events during the monitoring period. None of these events reached a critical severity level, indicating that although failed login attempts occurred, no major security incident was detected.

All authentication events observed during the selected timeframe were failed login attempts, with no successful authentications recorded. The Threat Hunting dashboard mapped these events to MITRE ATT&CK techniques such as Password Guessing, SSH Activity, and Brute Force, with Password Guessing representing the largest category.

Additional searches focused on SSH-related activity and produced results consistent with the authentication failure findings, confirming that the failed attempts were primarily associated with SSH access. Broader authentication searches returned similar results, validating the accuracy of the collected data.

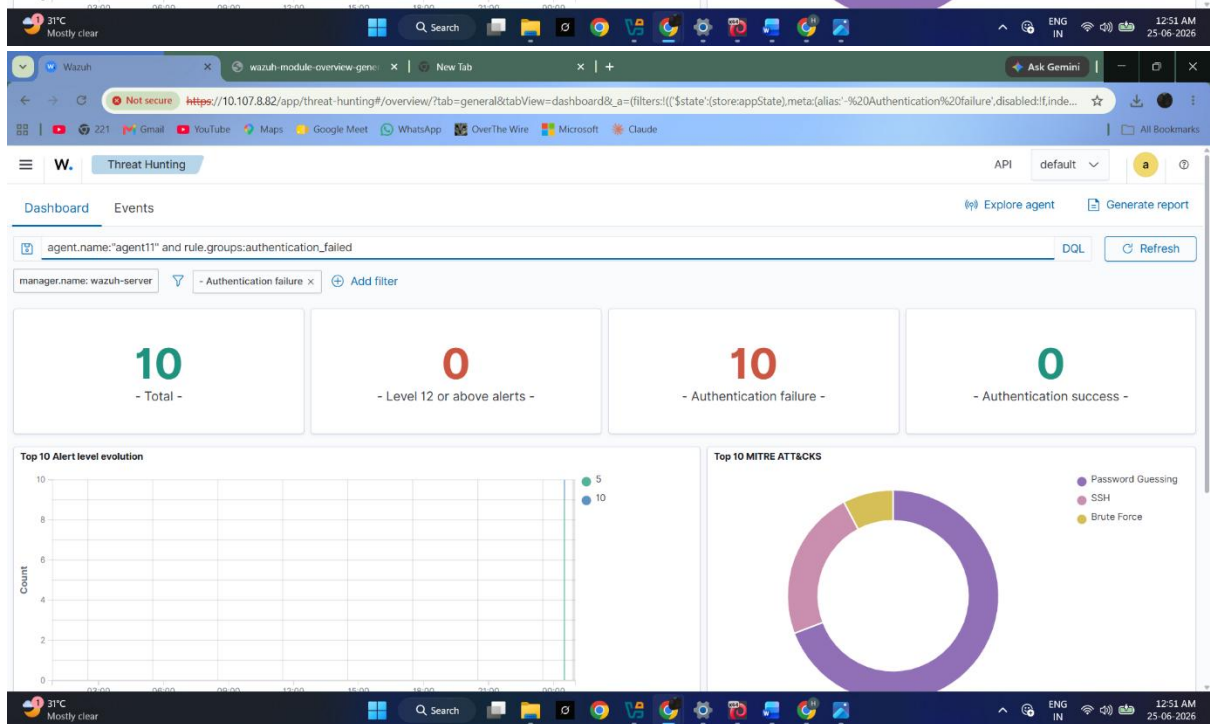
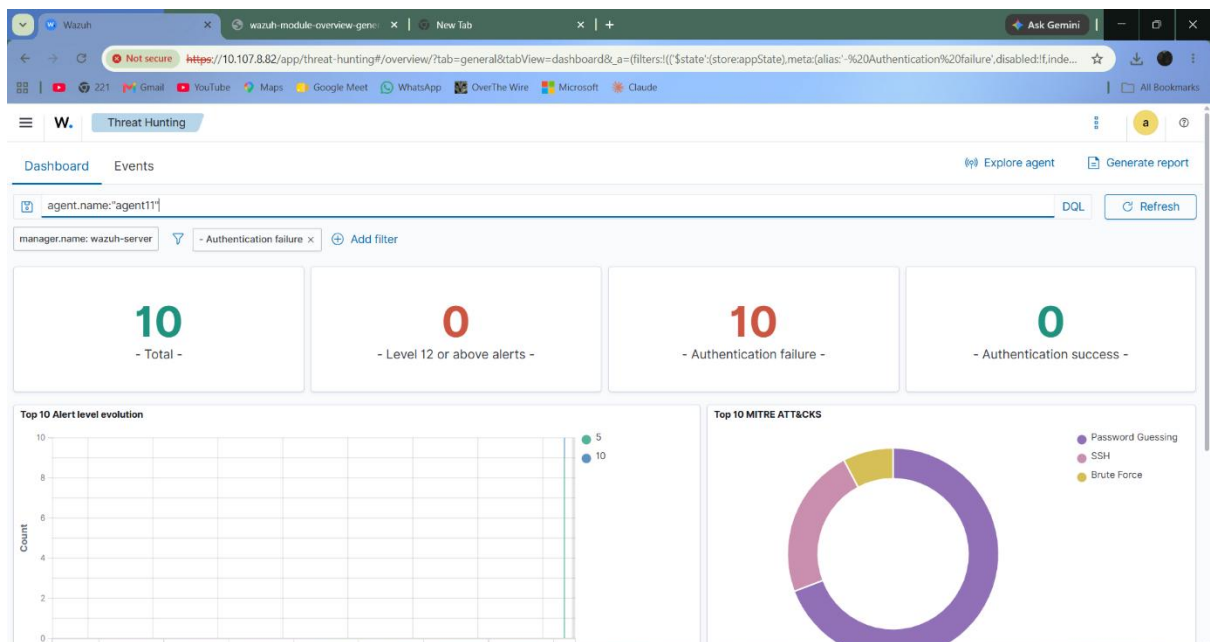
Beyond authentication monitoring, the event stream included rootcheck alerts, Security Configuration Assessment findings, CIS benchmark results, PAM session activity, and sudo-related events. These findings demonstrated the variety of security information collected from a single Linux endpoint and highlighted the value of filtering and threat hunting capabilities within Wazuh.

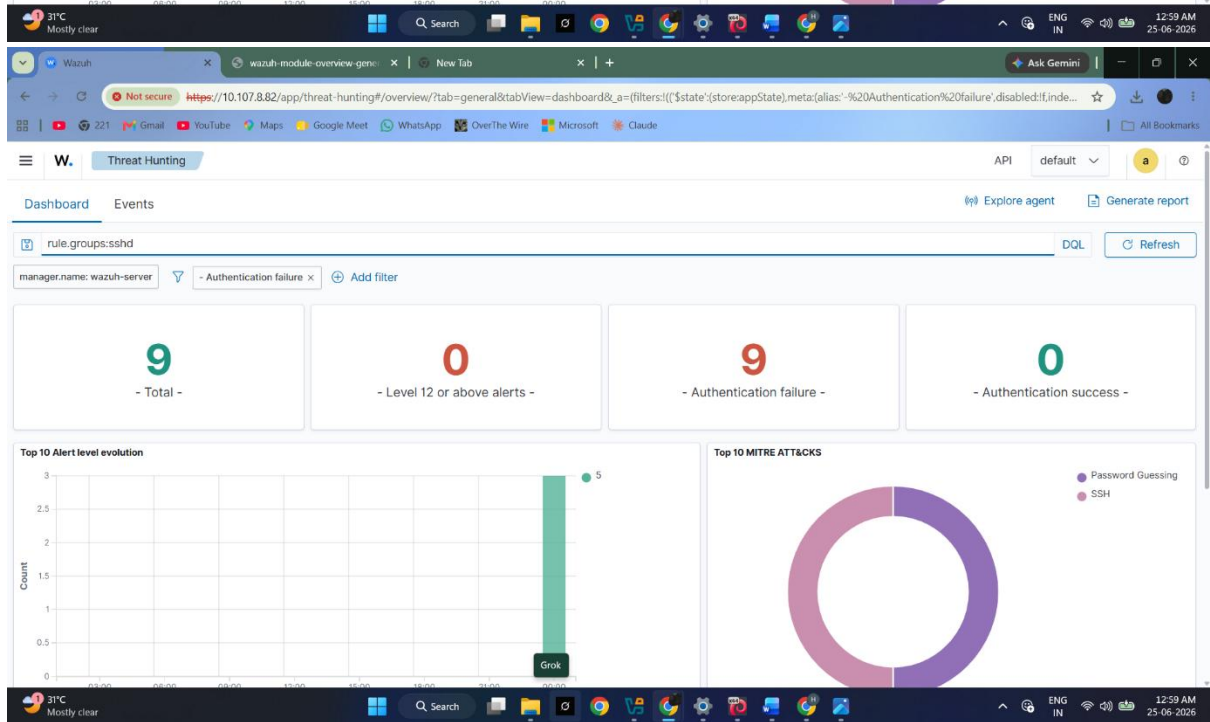
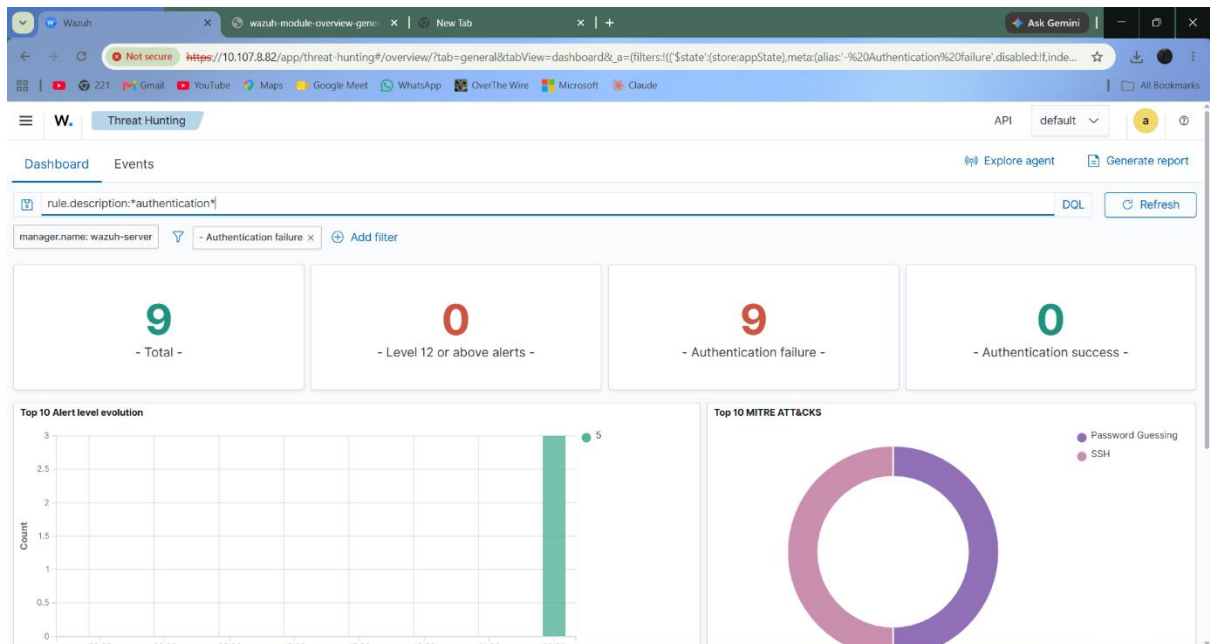
Recommendations

The authentication failures observed during this exercise should be monitored to determine their source and frequency. Understanding whether they originate from a single source or multiple sources would help distinguish between normal user errors and potential brute-force activity.

Implementing custom alert rules and active response actions for repeated authentication failures would strengthen the monitoring environment. Automated actions, such as temporarily blocking suspicious IP addresses after multiple failed login attempts, could improve protection against unauthorized access attempts.

The Security Configuration Assessment findings should also be reviewed and addressed to improve system hardening and overall security posture. As additional agents are added, creating dedicated dashboards and saved searches for specific agent groups will improve visibility and simplify security investigations.





W. Threat Huntingagent11APIdefaulta

Jun 24, 2026 @ 01:08:53.445 - Jun 25, 2026 @ 01:08:53.445

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timestamp	agent.name	rule.description	rule.level	rule.id
Jun 25, 2026 @ 01:05:53.8...	agent11	Host-based anomaly detection event (rootcheck).	7	510
Jun 25, 2026 @ 01:05:53.7...	agent11	Host-based anomaly detection event (rootcheck).	7	510
Jun 25, 2026 @ 01:05:53.7...	agent11	Host-based anomaly detection event (rootcheck).	7	510
Jun 25, 2026 @ 01:05:53.7...	agent11	Host-based anomaly detection event (rootcheck).	7	510
Jun 25, 2026 @ 01:05:53.7...	agent11	Host-based anomaly detection event (rootcheck).	7	510
Jun 25, 2026 @ 00:58:43.4...	agent11	SCA summary: CIS Distribution Independent Linux Benchmark v2.0.0.: Score less than 50% (47)	7	19004
Jun 25, 2026 @ 00:58:35.7...	agent11	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure rsyslog default file permissions configured.: Statu...	3	19010
Jun 25, 2026 @ 00:58:35.7...	agent11	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure rsyslog Service is enabled.: Status changed from f...	3	19010
Jun 25, 2026 @ 00:58:35.7...	agent11	CIS Distribution Independent Linux Benchmark v2.0.0.: Ensure rsyslog is installed.: Status changed from failed to...	3	19010
Jun 25, 2026 @ 00:58:28.0...	agent11	PAM: Login session opened.	3	5501
Jun 25, 2026 @ 00:58:20.0...	agent11	PAM: Login session opened.	3	5501
Jun 25, 2026 @ 00:58:20.0...	agent11	Successful sudo to ROOT executed.	3	5402
Jun 25, 2026 @ 00:58:09.7...	agent11	Host-based anomaly detection event (rootcheck).	7	510
Jun 25, 2026 @ 00:58:09.7...	agent11	Host-based anomaly detection event (rootcheck).	7	510

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WEEK 3: Attack Simulation & Detection

Executive Summary

Week 3 was about flipping roles for a bit and acting like the attacker instead of just watching the dashboard. I ran three small simulations on the Kali agent, repeated wrong password attempts to trigger brute force detection, dropped an EICAR test file to see if anything in Wazuh would pick it up, and ran a few basic commands to check whether that activity actually shows up in the logs. The point was to see if the manager actually reacts to real activity instead of just sitting there showing green.

Methodology

For the brute force part I just tried logging in with a handful of wrong passwords over SSH on purpose and then went back into Threat Hunting filtered on agent11 to see what got picked up. For the EICAR test I downloaded the standard EICAR test file onto the Kali agent and watched the syscheck related events to see if file integrity monitoring flagged it as something happening on disk. For command monitoring I ran a few basic commands locally, things like sudo activity and checking running processes, then pulled up the raw event logs for agent11 and searched through the ossec log output to see if that activity got captured.

Technical Findings

The brute force attempts generated 26 total events tied to authentication failure, with 0 reaching level 12 but one event specifically hitting level 10, a syslog alert for the user missing the password more than once, which is right in line with what the task was checking for. The rest of the failures were a mix of sshd authentication failed entries, unix_chkpwd password check failed entries, and PAM user login failed entries, all sitting around level 5. So the brute force attempt was clearly detected, just not severe enough on its own to cross into the level 12 critical range.

For the EICAR file, Wazuh didn't flag it through a dedicated malware or antivirus module since that wasn't configured, but file integrity monitoring picked up the activity around it, file added to the system, an integrity checksum change, and later a file deleted event. Those were tagged under MITRE ATT&CK as Data Destruction and File Deletion. So the detection came through FIM noticing the file appear and disappear rather than any signature based malware scan recognizing the EICAR string itself. Command monitoring activity showed up clearly in the raw logs, sudo to root being executed successfully, PAM login sessions opening and closing, and a ps command being run to list processes for all users, all logged with timestamps and rule IDs. So general command execution and session activity is definitely being captured, even if it's coming through as PAM and process related rules rather than something that explicitly labels it as whoami.

Recommendations

Since the brute force attempts only hit level 5 to 10 and never crossed into critical territory, it would help to either lower the alert threshold for repeated authentication failures or add a custom rule that escalates severity after a certain number of failed attempts in a short window. For malware detection, actually enabling a proper module like VirusTotal integration or a dedicated antivirus agent would give real detection instead of relying on FIM to indirectly notice file changes. For command monitoring, setting up a more specific rule or log source for shell command history would make it easier to search for exact commands run instead of inferring activity from PAM and process events alone.

The screenshot displays the Wazuh Threat Hunting interface in a web browser. The top section shows a table of events for the period Jun 24, 2026 @ 02:52:38.664 - Jun 25, 2026 @ 02:52:38.664. The table has columns for timestamp, agent.name, rule.description, rule.level, and rule.id. The events include PAM login sessions, successful sudo commands, and wazuh-agent status changes.

timestamp	agent.name	rule.description	rule.level	rule.id
Jun 25, 2026 @ 02:50:05.5...	agent11	PAM: Login session closed.	3	5502
Jun 25, 2026 @ 02:50:05.5...	agent11	PAM: Login session opened.	3	5501
Jun 25, 2026 @ 02:50:05.5...	agent11	Successful sudo to ROOT executed.	3	5402
Jun 25, 2026 @ 02:50:01.4...	wazuh-server	Listened ports status (netstat) changed (new port opened or closed).	7	533
Jun 25, 2026 @ 02:49:53.3...	agent11	Processes running for all users were queried with ps command.	6	92604
Jun 25, 2026 @ 02:49:32.2...	agent11	PAM: Login session closed.	3	5502
Jun 25, 2026 @ 02:49:32.1...	agent11	PAM: Login session opened.	3	5501
Jun 25, 2026 @ 02:49:32.1...	agent11	Successful sudo to ROOT executed.	3	5402
Jun 25, 2026 @ 02:49:26.2...	agent11	PAM: Login session closed.	3	5502
Jun 25, 2026 @ 02:49:18.1...	agent11	Wazuh agent started.	3	503
Jun 25, 2026 @ 02:49:16.0...	agent11	Wazuh agent stopped.	3	506
Jun 25, 2026 @ 02:48:54.6...	agent11	PAM: Login session closed.	3	5502
Jun 25, 2026 @ 02:48:54.6...	agent11	Successful sudo to ROOT executed.	3	5402
Jun 25, 2026 @ 02:48:54.6...	agent11	PAM: Login session opened.	3	5501
Jun 25, 2026 @ 02:48:44.6...	agent11	PAM: Login session closed.	3	5502

The bottom section shows a table of events for the period Jun 24, 2026 @ 01:51:06.606 - Jun 25, 2026 @ 01:51:06.606. The table has columns for timestamp, agent.name, rule.description, rule.level, and rule.id. The events include syslog password misses, sshd authentication failures, and unix_chkpwd password check failures.

timestamp	agent.name	rule.description	rule.level	rule.id
Jun 25, 2026 @ 01:50:59.0...	agent11	syslog: User missed the password more than one time	10	2502
Jun 25, 2026 @ 01:50:57.0...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:50:57.0...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:50:57.0...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:50:57.0...	agent11	unix_chkpwd: Password check failed.	5	5557
Jun 25, 2026 @ 01:50:55.0...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:50:55.0...	agent11	unix_chkpwd: Password check failed.	5	5557
Jun 25, 2026 @ 01:50:55.0...	agent11	unix_chkpwd: Password check failed.	5	5557
Jun 25, 2026 @ 01:50:55.0...	agent11	unix_chkpwd: Password check failed.	5	5557
Jun 25, 2026 @ 01:50:55.0...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:50:55.0...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:50:55.0...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:50:53.0...	agent11	PAM: User login failed.	5	5503
Jun 25, 2026 @ 01:50:53.0...	agent11	unix_chkpwd: Password check failed.	5	5557
Jun 25, 2026 @ 01:50:53.0...	agent11	PAM: User login failed.	5	5503

W. Threat Hunting agent11

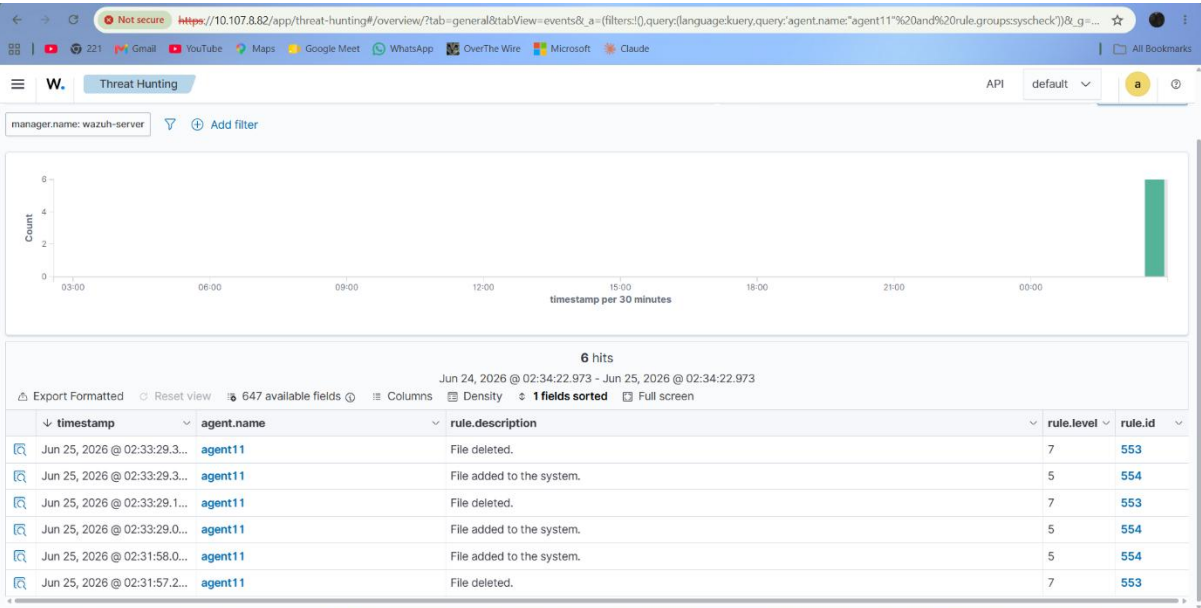
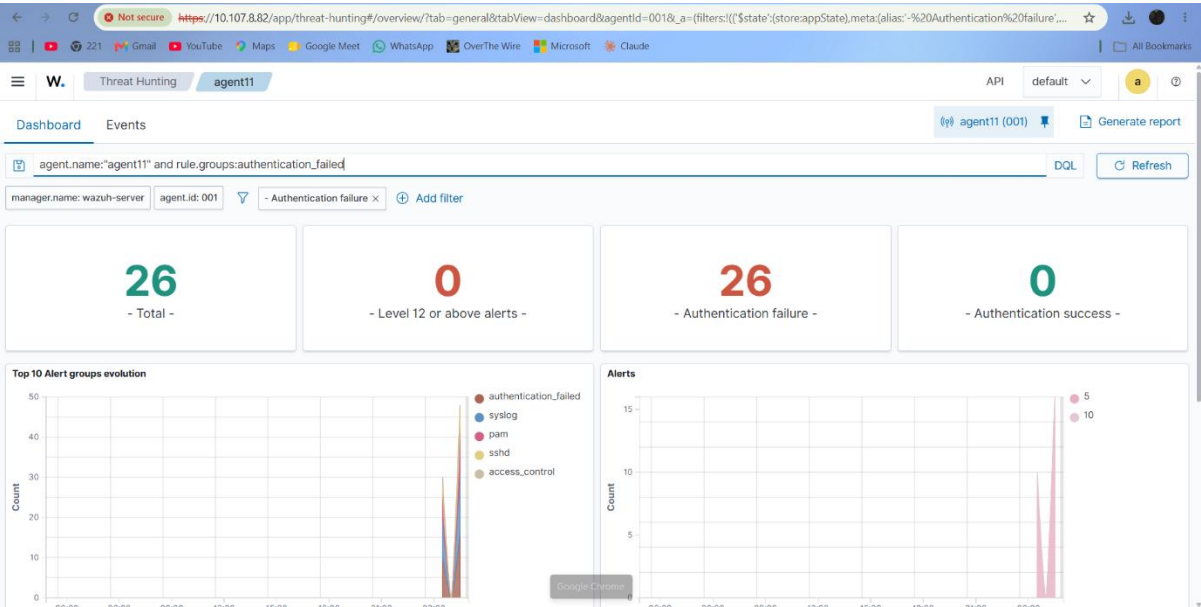
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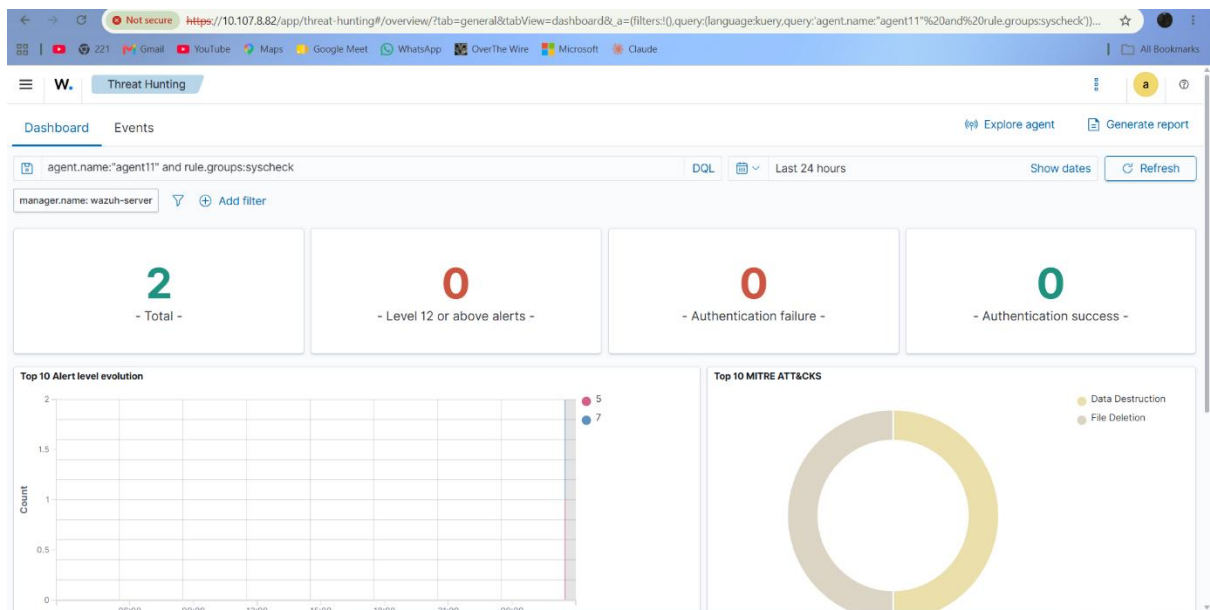
Jun 24, 2026 @ 01:49:54.923 - Jun 25, 2026 @ 01:49:54.923

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timestamp	agent.name	rule.description	rule.level	rule.id
Jun 25, 2026 @ 01:39:34.7...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:39:32.8...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:39:32.8...	agent11	unix_chkpwd: Password check failed.	5	5557
Jun 25, 2026 @ 01:39:30.8...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:39:30.8...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:39:30.8...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:39:30.7...	agent11	unix_chkpwd: Password check failed.	5	5557
Jun 25, 2026 @ 01:39:28.7...	agent11	sshd: authentication failed.	5	5760
Jun 25, 2026 @ 01:39:28.7...	agent11	PAM: User login failed.	5	5503
Jun 25, 2026 @ 01:39:28.7...	agent11	unix_chkpwd: Password check failed.	5	5557
Jun 25, 2026 @ 01:39:28.7...	agent11	PAM: User login failed.	5	5503
Jun 25, 2026 @ 01:39:28.7...	agent11	PAM: User login failed.	5	5503
Jun 25, 2026 @ 01:39:28.7...	agent11	unix_chkpwd: Password check failed.	5	5557
Jun 25, 2026 @ 01:39:28.7...	agent11	unix_chkpwd: Password check failed.	5	5557
Jun 25, 2026 @ 01:39:28.7...	agent11	PAM: User login failed.	5	5503

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Threat Hunting

API default

Jun 24, 2026 @ 02:50:50.033 - Jun 25, 2026 @ 02:50:50.033

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timestamp	agent.name	rule.description	rule.level	rule.id
Jun 25, 2026 @ 02:43:47.5...	agent11	File deleted.	7	553
Jun 25, 2026 @ 02:43:47.5...	agent11	File added to the system.	5	554
Jun 25, 2026 @ 02:43:47.4...	agent11	File deleted.	7	553
Jun 25, 2026 @ 02:43:47.4...	agent11	Integrity checksum changed.	7	550
Jun 25, 2026 @ 02:43:47.3...	agent11	File added to the system.	5	554
Jun 25, 2026 @ 02:41:55.4...	agent11	File added to the system.	5	554
Jun 25, 2026 @ 02:41:55.4...	agent11	File deleted.	7	553
Jun 25, 2026 @ 02:41:55.2...	agent11	File deleted.	7	553
Jun 25, 2026 @ 02:41:55.2...	agent11	File added to the system.	5	554
Jun 25, 2026 @ 02:40:45.4...	agent11	File deleted.	7	553
Jun 25, 2026 @ 02:40:45.4...	agent11	Integrity checksum changed.	7	550
Jun 25, 2026 @ 02:40:45.4...	agent11	File added to the system.	5	554
Jun 25, 2026 @ 02:40:38.1...	agent11	File deleted.	7	553
Jun 25, 2026 @ 02:40:38.1...	agent11	Integrity checksum changed.	7	550
Jun 25, 2026 @ 02:40:38.0...	agent11	File added to the system.	5	554

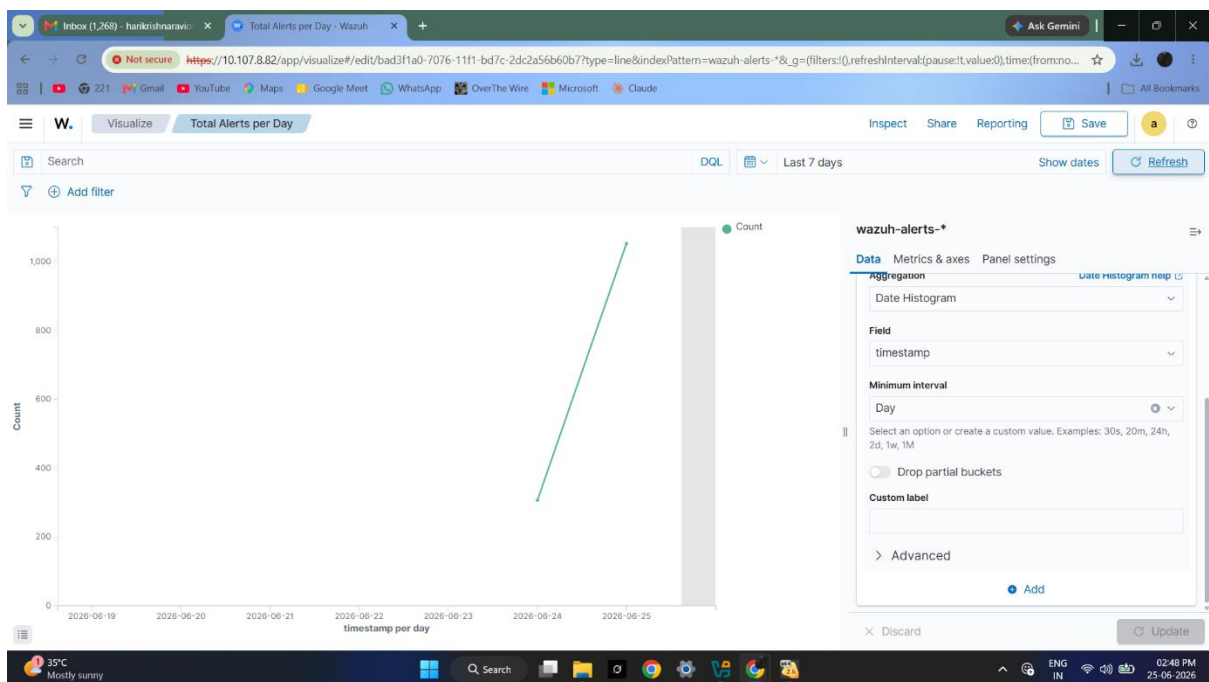
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WEEK 4: Dashboarding & Final Reporting

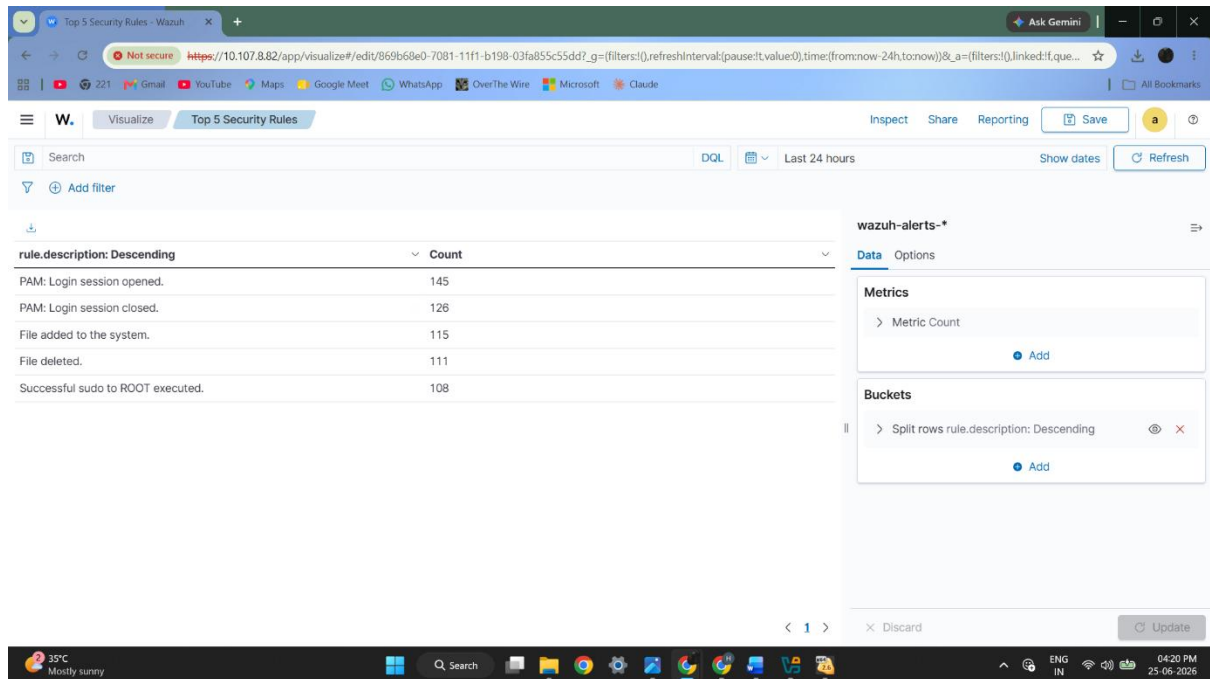
Objective: Presenting data like a Professional SOC Analyst.

Task 1: Custom Dashboard in Wazuh

Total Alerts per Day — Used wazuh-alerts index, created a bar chart with date histogram on the timestamp field to track daily alert volume.



Task 2: Top 5 Security Rules Triggered — Built a pie chart using rule.description field with Terms aggregation limited to 5 buckets to show most fired rules.



Task 3: Agent Status — Used wazuh-monitoring index, created a metric visualization splitting by status field to display Active, Disconnected, and Pending agents.

